

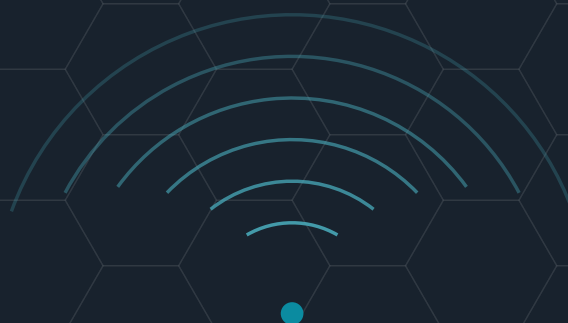
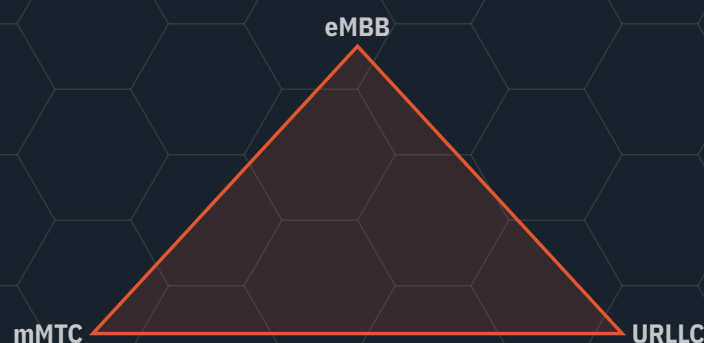
Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 14

5G Networks Introduction

مقدمه‌ای بر شبکه‌های 5G



Lecture Notes

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1 Why 5G? — Limitations of LTE/4G

Despite the success of LTE (4G), several fundamental limitations drove the need for a new generation:

1.1 Latency Floor (~10 ms)

- LTE's Transmission Time Interval (TTI) is fixed at **1 ms** subframe
- With scheduling, HARQ, and processing delays, the practical **round-trip latency floor is ~10 ms**
- This is insufficient for:
 - Autonomous vehicle control (requires < 5 ms)
 - Remote surgery / haptic feedback (requires ~1 ms)
 - Industrial real-time control loops

1.2 Capacity Ceiling

- LTE peak: ~1 Gbps (Cat-16, 3CA, 4×4 MIMO, 256-QAM) — theoretical
- Practical throughput: 100–300 Mbps in ideal conditions
- Spectrum below 6 GHz is congested; limited bandwidth per carrier (20 MHz max)
- 4×4 MIMO is the practical antenna limit in LTE form factors

1.3 IoT Scale

- LTE was designed for **human-centric** broadband communication
- LTE-M / NB-IoT added IoT support, but:
 - Connection density limited to ~10⁴ devices/km²
 - Not natively designed for **10⁶ devices/km²** required for smart city/industrial IoT
 - Signaling overhead per device is too high for massive deployments

1.4 Summary of LTE Limitations

Parameter	LTE Capability	5G Target
Peak data rate (DL)	~1 Gbps	20 Gbps
User-plane latency	~10 ms	1 ms
Connection density	~10 ⁴ /km ²	10 ⁶ /km ²
Spectral efficiency	30 bps/Hz	90 bps/Hz
Mobility	350 km/h	500 km/h

2 5G Requirements – ITU IMT-2020 Targets

The International Telecommunication Union (ITU) defined **IMT-2020** as the framework for 5G with the following key performance indicators (KPIs):

2.1 Peak Data Rate

- **Downlink:** 20 Gbps
- **Uplink:** 10 Gbps
- Achieved through wider bandwidth (up to 400 MHz per carrier) + higher-order MIMO + 256-QAM

2.2 Latency

- **User-plane latency:** 1 ms (URLLC scenario)
- **Control-plane latency:** 20 ms (idle to connected)
- Enables real-time control applications previously impossible on cellular

2.3 Connection Density

- **10⁶ devices per km²** (mMTC scenario)
- 100× improvement over LTE
- Support for sensors, actuators, meters with minimal overhead

2.4 User-Experienced Data Rate

- **100 Mbps** available everywhere (cell edge, dense urban)
- **50 Mbps UL** everywhere
- Represents the “guaranteed minimum” user experience

2.5 Reliability

- 99.999% (five nines) for URLLC
- Packet delivery within latency budget with 10^{-5} error probability
- Critical for safety-of-life applications

2.6 Additional IMT-2020 Targets

KPI	Target
Area traffic capacity	10 Mbps/m ²
Network energy efficiency	100× improvement
Spectrum efficiency	3× over IMT-Advanced
Mobility	500 km/h

3 Three Pillars – 5G Use Cases

5G is defined by three fundamental service categories (use-case pillars):

■ 5G Use Case Triangle

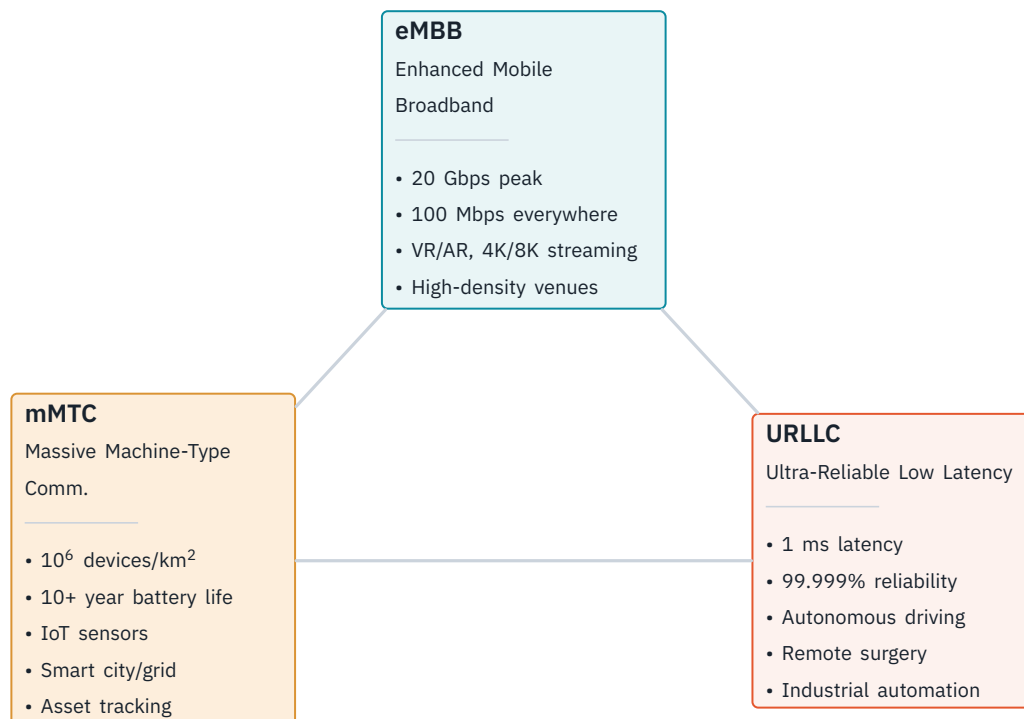


Figure 14.1 The 5G usage scenarios of ITU IMT-2020

3.1 eMBB – Enhanced Mobile Broadband

Focus: High data rates, capacity, coverage

Characteristic	Details
Peak rate	20 Gbps DL / 10 Gbps UL
User experience	100 Mbps everywhere
Key tech	Massive MIMO, mmWave, CA

Applications:

- Virtual Reality (VR) / Augmented Reality (AR) — requires >100 Mbps, <20 ms
- 4K/8K video streaming
- High-density event venues (stadiums: 100k+ users)
- Fixed Wireless Access (FWA) as fiber replacement

3.2 URLLC – Ultra-Reliable Low Latency Communications

Focus: Deterministic latency, extreme reliability

Characteristic	Details
Latency	1 ms (user plane, one way)
Reliability	99.999% (10^{-5} BLER)
Key tech	Mini-slots, pre-emption, redundancy

Applications:

- **Autonomous driving:** V2X communication, cooperative perception
- **Remote surgery:** Haptic feedback over cellular
- **Industry 4.0:** Real-time robotic control, motion control
- **Smart grid:** Protection and fault isolation

3.3 mMTC – Massive Machine-Type Communications

Focus: Scalability, energy efficiency, low cost

Characteristic	Details
Density	10^6 devices/km ²
Battery life	10–15 years on coin cell
Key tech	Grant-free access, NOMA, deep coverage

Applications:

- IoT sensor networks (environmental, agricultural)
- Smart city infrastructure (parking, waste, lighting)
- Smart metering (electricity, water, gas)
- Asset tracking and logistics
- Wearables

4 5G NR (New Radio) – Physical Layer

5G NR is the radio access technology defined in 3GPP Release 15+, designed from the ground up for flexibility.

4.1 Flexible Numerology

Unlike LTE (fixed 15 kHz SCS, 1 ms TTI), NR supports **multiple subcarrier spacings (SCS)**:

Numerology (μ)	SCS (kHz)	Slot Duration	CP Type	Typical Use
0	15	1 ms	Normal	FR1, eMBB (low freq)
1	30	0.5 ms	Normal	FR1, typical deployment
2	60	0.25 ms	Normal/Extended	FR1/FR2, URLLC
3	120	0.125 ms	Normal	FR2 (mmWave)
4	240	62.5 μ s	Normal	FR2 (SSB only)

Key relationships:

- Slot duration = $1 \text{ ms} / 2^\mu$
- Symbols per slot = 14 (normal CP) or 12 (extended CP)
- Higher SCS → shorter slot → lower latency (but wider subcarrier → more phase noise tolerance needed for mmWave)

4.2 Frequency Ranges

Range	Band	Frequencies	Max BW/carrier	Typical SCS
FR1	Sub-7 GHz	410 MHz – 7.125 GHz	100 MHz	15/30/60 kHz
FR2	mmWave	24.25 – 52.6 GHz	400 MHz	60/120/240 kHz

- **FR1** provides coverage and capacity (n77, n78 = C-band 3.3–4.2 GHz)
- **FR2** provides extreme capacity in short range (n257 = 28 GHz, n258 = 26 GHz, n260 = 39 GHz, n261 = 28 GHz)
- 3GPP Rel-17 introduced **FR2-2**: 52.6–71 GHz

4.3 Bandwidth Parts (BWP)

- A **Bandwidth Part** is a contiguous subset of the carrier bandwidth
- UE can be configured with up to **4 BWPs per serving cell** (1 active at a time)
- Allows UE power saving: narrow BWP for monitoring, wide BWP for data
- Each BWP can have its own numerology (SCS, CP)

4.4 Mini-Slots

- Standard NR slot = 14 symbols
- **Mini-slot** = 2, 4, or 7 OFDM symbols
- Enables sub-slot scheduling for **URLLC** without waiting for slot boundary
- Can **pre-empt** ongoing eMBB transmissions for urgent URLLC traffic

4.5 Massive MIMO and Beamforming

Feature	LTE	5G NR
Max antenna ports	16 (Rel-13 FD-MIMO)	256+
Beamforming	Fixed / semi-static	Dynamic, per-slot beam management
Beam management	N/A	SSB beams, CSI-RS, beam tracking
Panel architecture	Single panel	Multi-panel (FR2)

5G NR beam management procedures:

1. **P1**: Initial beam acquisition (SSB sweep)
2. **P2**: gNB Tx beam refinement (CSI-RS)
3. **P3**: UE Rx beam refinement
4. **Beam failure recovery**: Detect failure → select new candidate → RACH-based recovery

4.6 Key Differences from LTE PHY

Aspect	LTE	5G NR
Subcarrier spacing	Fixed 15 kHz	15/30/60/120/240 kHz
Max bandwidth	20 MHz	100 MHz (FR1) / 400 MHz (FR2)
Waveform DL	OFDMA	CP-OFDMA
Waveform UL	SC-FDMA	CP-OFDMA (+ DFT-s-OFDMA)
Reference signals	CRS always-on	No CRS; on-demand DMRS/CSI-RS
HARQ timing	Fixed (8 ms RTT)	Flexible, configurable
Scheduling	Slot-based only	Slot + mini-slot
Duplex	FDD or TDD (fixed)	Dynamic TDD, self-contained slot
Control channel	PDCCH in first 1-3 symbols	CORESET (flexible in time/freq)
DC subcarrier	Yes (LTE center)	No DC subcarrier

5 5G NSA vs SA – Deployment Strategies

5.1 Non-Standalone (NSA) – Option 3/3a/3x

Architecture: NR radio (gNB) + LTE core (EPC)

- The LTE eNB acts as the **master node** (anchor)
- NR gNB is the **secondary node** (adds capacity)
- Control plane goes through LTE (eNB → EPC)
- User plane can be split or switched

Sub-options:

- **Option 3:** User plane through eNB (split at eNB)
- **Option 3a:** User plane directly from gNB to EPC
- **Option 3x:** User plane split at gNB (gNB handles NR data + some LTE data)

5.2 Standalone (SA) – Option 2

Architecture: NR radio (gNB) + 5G Core (5GC)

- gNB connects directly to 5G Core via **NG interface**
- Full 5G functionality available
- No dependency on LTE infrastructure

5.3 Why NSA Was Deployed First

1. **Faster time-to-market:** Reuse existing EPC infrastructure
2. **Lower CAPEX:** No need to deploy 5G Core immediately
3. **Coverage anchor:** LTE provides ubiquitous coverage; NR adds throughput
4. **Proven core:** EPC is mature, well-understood
5. **Spectrum strategy:** Early NR deployments in limited bands

5.4 Why SA Is the Target

1. **Network Slicing:** Requires 5G Core SBA
2. **URLLC:** End-to-end low latency needs 5GC + MEC
3. **mMTC:** Native 5GC support for IoT optimizations
4. **Edge Computing:** UPF placement flexibility
5. **Independence:** No LTE dependency, cleaner architecture
6. **Service-Based Architecture:** Enables rapid service creation

5.5 NSA vs SA Architecture Comparison

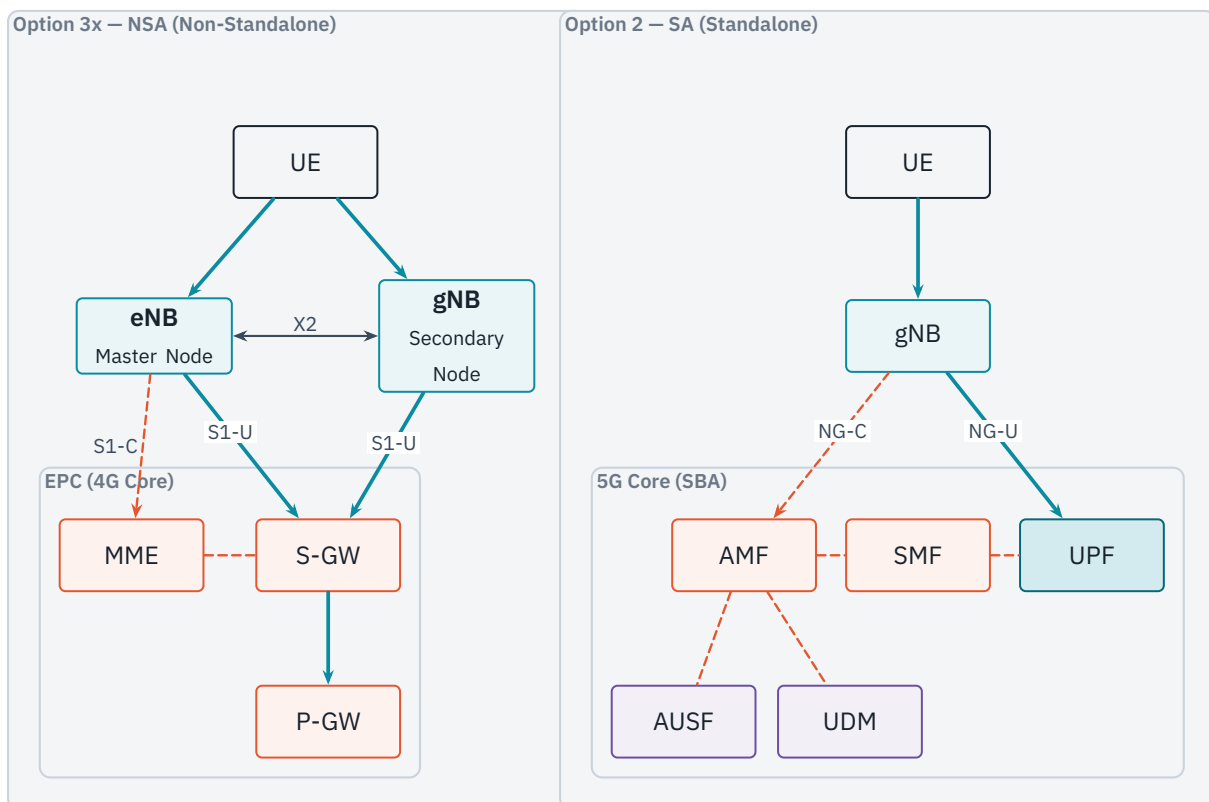


Figure 14.2 NSA Option 3x versus SA Option 2

5.6 Deployment Options Overview

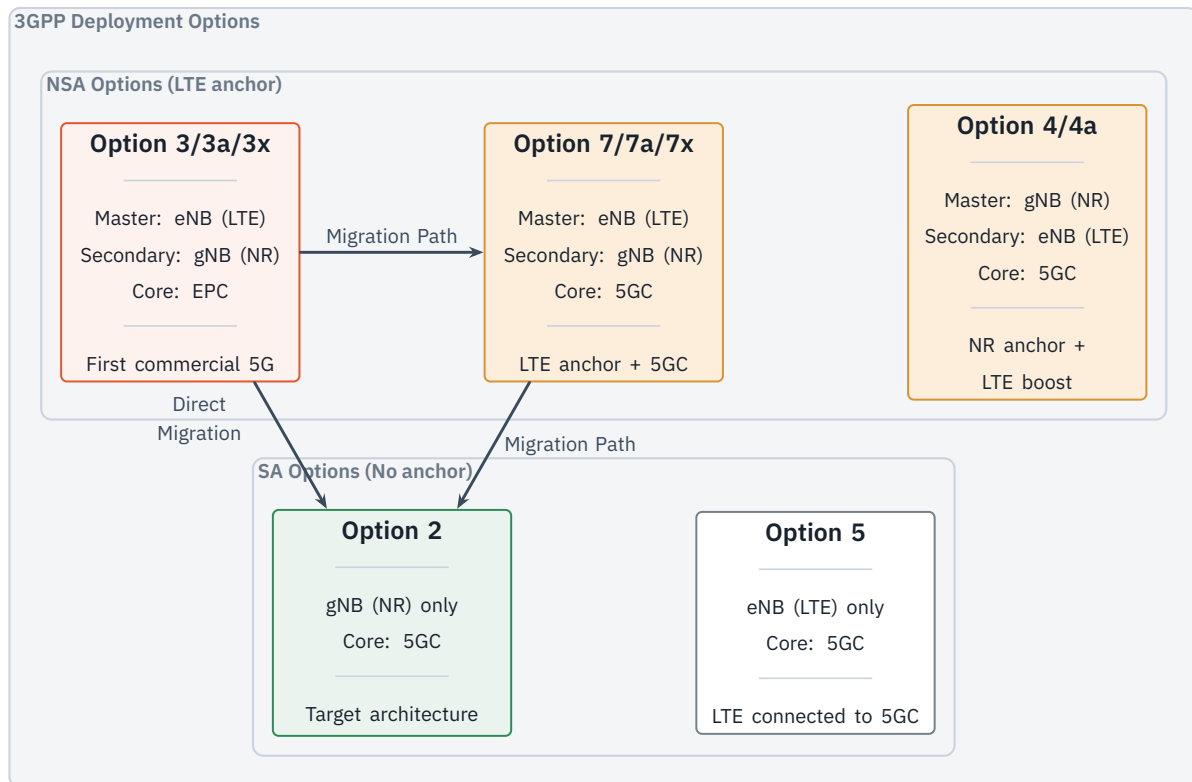


Figure 14.3 3GPP deployment options and migration paths

Migration path: Most operators follow **Option 3** → **Option 7** → **Option 2** or directly **Option 3** → **Option 2**.

6 5G Architecture Overview

6.1 NG-RAN (Next Generation Radio Access Network)

Node	Full Name	Description
gNB	Next Generation Node B	5G NR base station (SA)
en-gNB	E-UTRA-NR gNB	NR node in NSA (secondary to eNB)
ng-eNB	Next Generation eNB	LTE node connected to 5GC

Interfaces:

- **Xn:** Between gNBs (and ng-eNBs) — handover, dual connectivity
- **NG:** Between gNB and 5G Core (NG-C to AMF, NG-U to UPF)
- **F1:** Internal gNB split (CU ↔ DU)
- **E1:** CU-CP ↔ CU-UP

6.2 5G Core – Service-Based Architecture (SBA)

The 5G Core replaces the monolithic EPC with **microservices**:

Network Function	Role
AMF (Access & Mobility Mgmt)	Registration, connection, mobility, NAS security
SMF (Session Management)	PDU session establishment, QoS, UPF selection
UPF (User Plane Function)	Packet routing, inspection, QoS enforcement
AUSF (Authentication Server)	Authentication procedures
UDM (Unified Data Mgmt)	Subscription data, similar to HSS
PCF (Policy Control)	Policy rules, replaces PCRF
NSSF (Network Slice Selection)	Slice selection and management
NEF (Network Exposure)	API exposure to external applications
NRF (NF Repository)	Service discovery and registration

Key SBA principles:

- NFs communicate via **HTTP/2 REST APIs** (service-based interfaces)
- **Stateless** design: compute and storage separated
- **Cloud-native**: containerized, horizontally scalable
- **Service discovery**: NRF enables dynamic NF registration

6.3 Separation of Radio Technology from Network Architecture

A critical 5G design principle:

Radio Access Technology (RAT) $\neq$ Core Network Architecture

- LTE radio can connect to 5G Core (ng-eNB → Option 5/7)
- NR radio can connect to EPC (en-gNB → Option 3)
- **Decoupling** enables flexible migration and coexistence

7 gNB Architecture – CU/DU Split

7.1 Functional Split

The gNB can be decomposed into:

Component	Function	Location
CU (Central Unit)	RRC, SDAP, PDCP	Centralized (edge DC)
DU (Distributed Unit)	RLC, MAC, High-PHY	Near antenna site
RU (Radio Unit)	Low-PHY, RF	At antenna

7.2 CU-CP / CU-UP Split

The CU is further split:

- **CU-CP** (Control Plane): RRC + control part of PDCP
- **CU-UP** (User Plane): SDAP + user part of PDCP

This enables:

- Independent scaling of control and user plane
- Multiple CU-UPs per CU-CP
- Flexible placement (CU-UP closer to edge for low latency)

7.3 gNB CU/DU Split Architecture

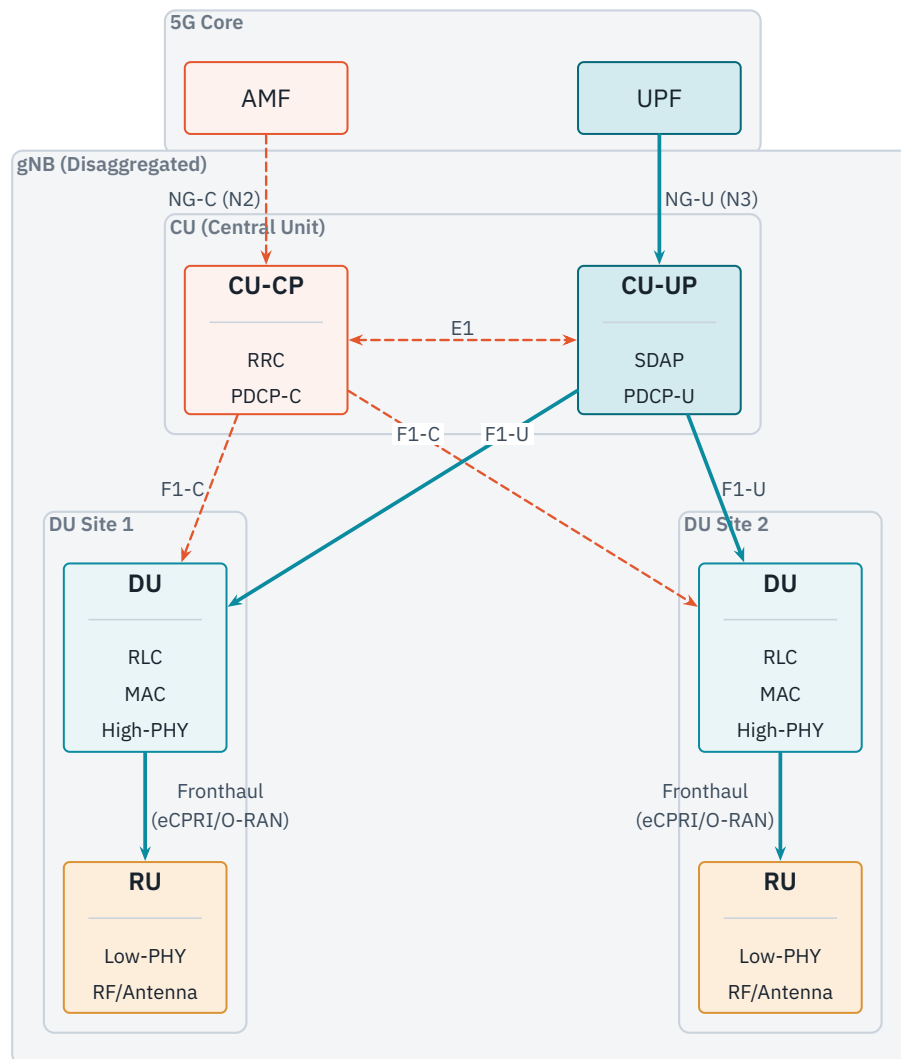


Figure 14.4 Disaggregated gNB: CU-CP, CU-UP, DU and RU

7.4 Interface Summary

Interface	Endpoints	Protocol	Purpose
NG-C (N2)	gNB ↔ AMF	NGAP/SCTP	Control plane signaling
NG-U (N3)	gNB ↔ UPF	GTP-U/UDP	User data transport
Xn-C	gNB ↔ gNB	XnAP/SCTP	Handover, dual connectivity
Xn-U	gNB ↔ gNB	GTP-U/UDP	Data forwarding during HO
F1-C	CU-CP ↔ DU	F1AP/SCTP	RRC message transport, UE context
F1-U	CU-UP ↔ DU	GTP-U/UDP	User data between CU-UP and DU
E1	CU-CP ↔ CU-UP	E1AP/SCTP	Bearer context management

7.5 Deployment Scenarios

Scenario	CU Location	DU Location	Fronthaul	Use Case
Co-located	Cell site	Cell site	Internal	Rural, simple
Centralized CU	Edge DC	Cell site	Midhaul (F1)	Urban, pooling
Full disaggregation	Regional DC	Aggregation	Mid + Fronthaul	Cloud RAN, O-RAN

8 Timeline – 3GPP Releases

8.1 Release Overview

Release	Freeze Date	Key Features
Rel-15	2018 (Late Drop: 2019)	First 5G NR specification; NSA & SA; eMBB focus
Rel-16	2020 (Q3)	URLLC enhancements; V2X; NR-U (unlicensed); IIoT; NR positioning
Rel-17	2022 (Q1)	FR2-2 (52-71 GHz); RedCap (reduced capability); NTN (satellite); NR sidelink; MBS
Rel-18	2024 (Q1)	5G-Advanced; AI/ML for air interface; XR optimization; duplex evolution; network energy saving

8.2 Evolution Path

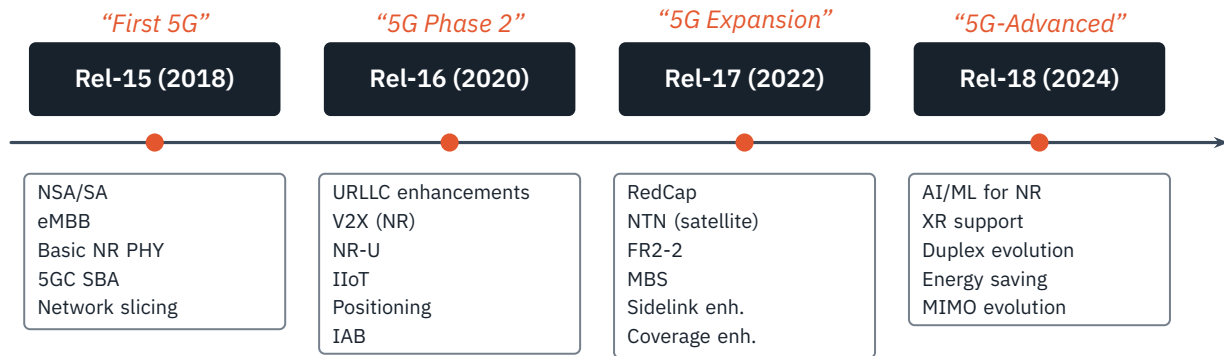


Figure 14.5 Evolution path from Release 15 to Release 18

8.3 Key Milestones

- **Dec 2017:** Rel-15 NSA (Option 3) specification completed (early drop)
- **Jun 2018:** Rel-15 SA (Option 2) specification completed
- **Apr 2019:** First commercial 5G NSA networks launched (South Korea, US)
- **2020:** First 5G SA commercial deployments (T-Mobile US, China operators)
- **2022:** Rel-17 freeze; RedCap enables mid-tier IoT on NR
- **2024:** Rel-18 (5G-Advanced) freeze; bridge to 6G research

Summary – Key Takeaways

1. **5G addresses three fundamental LTE limitations:** latency, capacity, and IoT scale
2. **IMT-2020 targets** define quantitative goals: 20 Gbps, 1 ms, 10^6 devices/km²
3. **Three pillars** (eMBB/URLLC/mMTC) serve different verticals with different KPIs
4. **5G NR** introduces flexible numerology, mmWave support, and dynamic beamforming
5. **NSA enables quick deployment** by reusing LTE/EPC; **SA unlocks full 5G** potential
6. **SBA-based 5G Core** is cloud-native, microservice-based, enabling slicing and edge
7. **gNB disaggregation** (CU/DU/RU) enables flexible, scalable deployments
8. **3GPP releases** progressively add features: 15→basic, 16→URLLC, 17→expansion, 18→5G-Advanced

References

- 3GPP TS 38.300: NR and NG-RAN Overall Description
- 3GPP TS 38.401: NG-RAN Architecture Description
- 3GPP TS 23.501: System Architecture for 5G System
- ITU-R M.2083: IMT Vision — Framework and Overall Objectives of the Future Development of IMT for 2020 and Beyond
- 3GPP TS 38.211: NR Physical Channels and Modulation
- 3GPP TS 38.104: NR Base Station Radio Transmission and Reception

Module 14 — Advanced Communication Networks Course

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